preceding paragraph in that both the challenge and response are derived in real-time.

The strength of challenge-response systems is that they use artifacts derived in real-time, making it nearly impossible to reuse the response, even if it's intercepted. This strategy protects against phishing and replay attacks, so it is often used in cryptographic protocols, device authentication, and phishing-resistant flows.

DEFINITION

A *replay attack* occurs when an attacker captures a valid authentication message—like a one-time code sent via SMS—and reuses it to gain unauthorized access. If the system doesn't verify that the message is associated with a unique challenge, it may accept the replayed code. Challenge-response protocols prevent this by ensuring each authentication request has a distinct, non-reusable challenge.

To see how this can work, imagine that the authentication service is using public-private key cryptography to create the responses as shown in figure 3.8. Compare figure 3.8 with figure 3.7 and you'll notice the flows are quite different.